2026-W46 / 2026-11-09

Atlas modernization assessment

Feasibility, effort & migration worthiness

SYNTHETIC DEMONSTRATION · All measurements, costs and compatibility observations are illustrative. This report tracks assessment research, not migration delivery.

Executive Overview

ANALYSIS COVERAGE	EVIDENCE VALIDATED	OPEN UNCERTAINTIES	DECISION GATES
74.2%	**2** / 8	**3**	**2** / 4
Examined share of planned research scope	Validated evidence / all evidence	Unanswered research questions	Met gates; critical unknowns still block

Current assessment

defer

Defer migration commitment; continue focused assessment.

Technical feasibility is increasingly evidenced, but worthiness is not established without the economic counterfactual and recovery assessment.

recommended · 2 critical gates outstanding

- Unknown benefits
- No recovery evidence
- License exception unresolved

□ F-API · □ F-ADAPTER · □ F-DEPS

What management needs to know

CURRENT SYNTHESIS

- The week changed our understanding: namespace-only session compatibility was contradicted; an explicit adapter is now evidenced on synthetic fixtures.
- Representative transaction evidence strengthened, and dependency alternatives partially matured. Technical adaptation effort rises to 95–155 person-days.
- Defer migration commitment: benefits remain unknown and operational recovery is insufficiently assessed. Next priorities are the economic counterfactual, recovery experiment and license position.

> Research coverage describes what has been examined. It does not measure confidence, feasibility, or migration completion.

Assessment dimensions

Framework compatibility 100%

Examined scope 3 / 3 · dimension weight 25

Dependency viability 100%

Examined scope 3 / 3 · dimension weight 20

Effort & worthiness 33.3%

Examined scope 1 / 3 · dimension weight 20

Operational impact 0%

Examined scope 0 / 1 · dimension weight 5

Runtime & platform 50%

Examined scope 1 / 2 · dimension weight 15

Security & compliance 100%

Examined scope 2 / 2 · dimension weight 15

What changed our understanding

`F-ADAPTER`

Explicit session adapter required

added

`F-API`

Representative API adaptation validated

strengthened

`F-DEPS`

Dependency path partially corroborated

strengthened

`F-SERIAL`

Namespace-only session hypothesis contradicted

contradicted

> 2 critical gates require further evidence before a proceed recommendation.

Assessment Coverage

Planned scope & examined coverage

Framework compatibility 100%

Examined scope 3 / 3 · dimension weight 25

Dependency viability 100%

Examined scope 3 / 3 · dimension weight 20

Effort & worthiness 33.3%

Examined scope 1 / 3 · dimension weight 20

Operational impact 0%

Examined scope 0 / 1 · dimension weight 5

Runtime & platform 50%

Examined scope 1 / 2 · dimension weight 15

Security & compliance 100%

Examined scope 2 / 2 · dimension weight 15

How these measures are calculated

Coverage uses planned investigation scope weights within each dimension, then business-defined dimension weights. Preliminary or later outcomes count as examined; started work does not.

Validated evidence: 2/8. Corroborated or validated findings: 3/4. Rejected evidence and assumptions remain in the denominator.

No planned investigation scope means Unknown, not zero confidence or favorable evidence. An inconclusive outcome increases examined coverage while leaving uncertainty explicit.

Investigation register

INVESTIGATION / QUESTION	DIMENSION	LIFECYCLE	OWNER / NEXT INQUIRY	PROVENANCE
I-API **Are API and namespace adaptations bounded?**	Framework compatibility Scope weight 2	validated	Research team Validate a representative transaction slice.	
I-AUTH **Are authorization semantics equivalent?**	Security & compliance Scope weight 1	preliminary	Research team Exercise denied and cross-tenant requests.	
I-CAPACITY **Is performance acceptable at peak load?**	Runtime & platform Scope weight 1	not-examined	Research team Design a replay dataset without customer data.	
I-DEPS **Do maintained replacement libraries exist?**	Dependency viability Scope weight 2	corroborated	Research team Reconcile the full transitive inventory.	
I-EFFORT **What effort range is defensible?**	Effort & worthiness Scope weight 1	preliminary	Research team Model remediation sensitivity.	
I-LICENSE **Are replacement licenses acceptable?**	Dependency viability Scope weight 1	preliminary	Research team Request legal review of candidate licenses.	
I-RECOVERY **Can operations sustain the candidate platform?**	Operational impact Scope weight 1	investigating	Research team Interview operations and test restore assumptions.	
I-ROI **Are benefits and avoided costs worth the effort?**	Effort & worthiness Scope weight 2	investigating	Research team Validate the counterfactual and benefit horizon.	
I-RUNTIME **Does the target runtime host representative workloads?**	Runtime & platform Scope weight 1	preliminary	Research team Run a transaction and recovery experiment.	
I-SERIAL **Can stored serialized sessions be read?**	Framework compatibility Scope weight 1	corroborated	Research team Capture legacy session fixtures.	
I-SUPPORT **Is the support lifecycle adequate?**	Security & compliance Scope weight 1	inconclusive	Research team Obtain written support assumptions.	

Findings & Evidence

F-ADAPTER · revision 1

corroborated

Explicit session adapter required

An explicit serialization version adapter is required; it passes all sampled fixtures.

medium confidence mixed critical

This informs feasibility and estimated research needs; it is not implementation completion.

UNCERTAINTIES

- Production fixture diversity unmeasured

Next: Expand fixture diversity before accepting broad compatibility.

Supersedes F-SERIAL. Earlier conclusions remain in history.

□ E-SERIAL · validated

F-API · revision 2

validated

Representative API adaptation validated

The representative transaction slice passes all 18 rollback cases with the adapter layer.

high confidence favorable high

This informs feasibility and estimated research needs; it is not implementation completion.

UNCERTAINTIES

- Distributed recovery excluded

Next: Test failure recovery under restart.

□ E-TX · validated

F-DEPS · revision 2

corroborated

Dependency path partially corroborated

Two maintained replacement candidates pass smoke checks; one wrapper and license exception remain unresolved.

medium confidence mixed high

This informs feasibility and estimated research needs; it is not implementation completion.

UNCERTAINTIES

- License exception not approved
- Third wrapper untested

Next: Resolve license and isolation questions.

□ E-DEPS · reviewed □ E-DEPS2 · reviewed

F-SERIAL · revision 2

contradicted

Namespace-only session hypothesis contradicted

Nine of 32 synthetic legacy fixtures fail without an explicit serialization adapter. The prior assumption is contradicted.

low confidence adverse critical

This informs feasibility and estimated research needs; it is not implementation completion.

UNCERTAINTIES

- Legacy session diversity unknown

Next: Use F-ADAPTER for subsequent evaluation.

□ E-SERIAL · validated

Evidence register

EVIDENCE	VERIFICATION	KIND / LIMITATIONS	TRACE
E-ASSUME Assumed serialization equivalence after namespace adaptation.	unverified	assumption No fixture-based test exists.	□ E-ASSUME · unverified
E-DEPS Synthetic dependency inventory and candidate maintenance scan.	reviewed	analysis Only 80% of direct coordinates reconciled; transitive review pending.	□ E-DEPS · reviewed
E-DEPS2 Complete synthetic direct inventory with two interface smoke checks.	reviewed	analysis License exception unresolved; third wrapper experiment pending.	□ E-DEPS2 · reviewed
E-MODEL Synthetic effort scenario model with explicitly incomplete benefits.	reviewed	financial-model No calibrated productivity baseline, benefit horizon or operating-cost counterfactual.	□ E-MODEL · reviewed
E-MODEL2 Revised synthetic effort model incorporating session adapter work.	reviewed	financial-model Benefits, recovery and rollout remain uncalibrated; estimates are not commitments.	□ E-MODEL2 · reviewed
E-SCAN Synthetic static namespace/API inventory across six modules.	reviewed	analysis Generated code, persistence metadata and runtime behavior excluded.	□ E-SCAN · reviewed
E-SERIAL Synthetic legacy fixture experiment: 9/32 fail without adapter; 32/32 pass with adapter.	validated	experiment Synthetic fixture set; production diversity unmeasured.	□ E-SERIAL · validated
E-TX Synthetic transaction rollback experiment: 18/18 passed on a representative slice.	validated	experiment Sampled cases only; distributed recovery excluded.	□ E-TX · validated

Risks & Unknowns

Potential risks and compatibility concerns

RISK	LIKELIHOOD / IMPACT	STATUS / OWNER	POSSIBLE MITIGATION	EVIDENCE
R-SESSION **Stored session incompatibility raises adaptation effort and backward-compatibility uncertainty.**	high likelihood critical impact	open Runtime research lead	• Explicit version adapters • Investigate session invalidation trade-offs	□ E-SERIAL · validated
R-WRAPPERS **Unmaintained wrappers may require custom replacement and increase effort.**	medium likelihood high impact	open Integration research lead	• Evaluate maintained alternatives • Isolate through an adapter boundary	□ E-DEPS · reviewed

Uncertainties & unanswered questions

QUESTION	STATUS	OWNER / NEXT INVESTIGATION	TRACE
Q-LEGAL **Are candidate replacement licenses acceptable?** Decision-blocking question	open	Legal liaison Request a documented license position.	□ E-DEPS · reviewed
Q-ROI **What counterfactual cost and benefit horizon should management use?** Decision-blocking question	open	Finance partner Validate avoided-cost assumptions.	Unverified · no supporting evidence
Q-SERIAL **Will legacy session fixtures deserialize safely?** Decision-blocking question	answered	Runtime research lead Expand production-representative fixture diversity.	□ E-SERIAL · validated
Q-SUPPORT **Which support terms and security policy oracle are authoritative?** Decision-blocking question	open	Security & procurement Obtain written terms and an approved deny-policy oracle.	□ E-DEPS2 · reviewed

Alternatives & Trade-offs

> The preferred research alternative is not authorization to execute a migration. Unknown effort and unverified benefits stay explicit.

ALTERNATIVE	POTENTIAL BENEFITS	DRAWBACKS / ASSUMPTIONS	EFFORT / ASSESSMENT	EVIDENCE
ALT-REPLACE **Replace with a new service implementation**	• Potential simplification	• Functional parity and data migration unassessed Assumptions: Business processes can be re-specified	unassessed Not yet measurable	Unverified · no supporting evidence
ALT-RETAIN **Retain the current platform**	• No immediate conversion expenditure	• Unquantified support and opportunity costs Assumptions: Current support remains purchasable	viable Unknown; support stabilization not estimated	□ E-MODEL · reviewed
ALT-STAGED **Assess a staged modernization**	• Limits exposure while testing critical hypotheses	• Parallel-platform overhead remains unknown Assumptions: Representative slices predict key compatibility risks	preferred See EST-EFFORT; not a delivery commitment	□ E-SCAN · reviewed □ E-MODEL · reviewed

Effort, Cost & Benefit

Estimates are ranges with stated assumptions, not commitments. No ROI or combined total is calculated while benefits, horizons, or compatible units are missing.

Evidence-backed estimates

ESTIMATE	RANGE	CONFIDENCE	ASSUMPTIONS / EXCLUSIONS	EVIDENCE
EST-BENEFIT · benefit **Recurring economic benefit**	**Not yet measurable**	unknown	• Benefit horizon and counterfactual not yet agreed	Unverified · no supporting evidence
EST-COST · cost **One-time external assessment of transition costs**	**35,000–65,000** USD	low	• Synthetic quotes • Licensing renegotiation excluded	□ E-MODEL · reviewed
EST-EFFORT · effort **Technical adaptation effort**	**95–155** person-days	medium	• Explicit serialization adapter now required • Includes wrapper isolation uncertainty • Excludes rollout and unassessed recovery effort	□ E-MODEL2 · reviewed □ E-SERIAL · validated

What the economics currently establish

Technical feasibility is increasingly evidenced, but worthiness is not established without the economic counterfactual and recovery assessment.

- Unknown benefits
- No recovery evidence
- License exception unresolved

Weekly Evolution

28 record-level changes **1** contradicted conclusions **2** strengthened findings

`investigations / I-API · revision 1 → 2` revised

Are API and namespace adaptations bounded?

▶ 1 changed fields · inspect previous and current understanding

`investigations / I-DEPS · revision 1 → 2` revised

Do maintained replacement libraries exist?

▶ 1 changed fields · inspect previous and current understanding

`investigations / I-LICENSE · revision 1 → 2` revised

Are replacement licenses acceptable?

▶ 1 changed fields · inspect previous and current understanding

`investigations / I-RECOVERY · revision 1 → 2` revised

Can operations sustain the candidate platform?

▶ 1 changed fields · inspect previous and current understanding

`investigations / I-ROI · revision 1 → 2` revised

Are benefits and avoided costs worth the effort?

▶ 1 changed fields · inspect previous and current understanding

`investigations / I-RUNTIME · revision 1 → 2` revised

Does the target runtime host representative workloads?

▶ 1 changed fields · inspect previous and current understanding

investigations / I-SERIAL · revision 1 → 2

revised

Can stored serialized sessions be read?

▶ 1 changed fields · inspect previous and current understanding

investigations / I-SUPPORT · revision 1 → 2

revised

Is the support lifecycle adequate?

▶ 1 changed fields · inspect previous and current understanding

findings / F-ADAPTER · revision — → 1

added

Explicit session adapter required

New record. Its complete content is included in the cumulative report.

findings / F-API · revision 1 → 2

strengthened

Representative API adaptation validated

▼ 7 changed fields · inspect previous and current understanding

BEFORE · CONCLUSION

The sampled API surface suggests a manageable adapter layer; estate-wide extrapolation is premature.

NOW · CONCLUSION

The representative transaction slice passes all 18 rollback cases with the adapter layer.

BEFORE · CONFIDENCE

medium

NOW · CONFIDENCE

high

BEFORE · EVIDENCEREFS

["E-SCAN"]

NOW · EVIDENCEREFS

["E-TX"]

BEFORE · NEXTINVESTIGATION

Validate transaction slices.

NOW · NEXTINVESTIGATION

Test failure recovery under restart.

BEFORE · STATUS

preliminary

NOW · STATUS

validated

BEFORE · TITLE

API adaptation appears bounded

NOW · TITLE

Representative API adaptation validated

BEFORE · UNCERTAINTIES

["Generated code not scanned"]

NOW · UNCERTAINTIES

["Distributed recovery excluded"]

findings / F-DEPS · revision 1 → 2

strengthened

Dependency path partially corroborated

▼ 8 changed fields · inspect previous and current understanding

BEFORE · CONCLUSION

Three candidate wrappers lack a demonstrated maintenance path.

NOW · CONCLUSION

Two maintained replacement candidates pass smoke checks; one wrapper and license exception remain unresolved.

BEFORE · CONFIDENCE

low

NOW · CONFIDENCE

medium

BEFORE · EVIDENCEREFS

["E-DEPS"]

NOW · EVIDENCEREFS

["E-DEPS","E-DEPS2"]

BEFORE · IMPACT

adverse

NOW · IMPACT

mixed

BEFORE · NEXTINVESTIGATION

Investigate maintained replacements.

NOW · NEXTINVESTIGATION

Resolve license and isolation questions.

BEFORE · STATUS

preliminary

NOW · STATUS

corroborated

BEFORE · TITLE

Unmaintained integration wrappers

NOW · TITLE

Dependency path partially corroborated

BEFORE · UNCERTAINTIES

["Replacement license costs unknown"]

NOW · UNCERTAINTIES

["License exception not approved","Third wrapper untested"]

findings / F-SERIAL · revision 1 → 2

contradicted

Namespace-only session hypothesis contradicted

▼ 8 changed fields · inspect previous and current understanding

BEFORE · CONCLUSION

Namespace-only adaptation is assumed to preserve serialized session behavior.

NOW · CONCLUSION

Nine of 32 synthetic legacy fixtures fail without an explicit serialization adapter. The prior assumption is contradicted.

BEFORE · CONFIDENCE

high

NOW · CONFIDENCE

low

BEFORE · EVIDENCEREFS

["E-ASSUME"]

NOW · EVIDENCEREFS

["E-SERIAL"]

BEFORE · IMPACT

favorable

NOW · IMPACT

adverse

BEFORE · NEXTINVESTIGATION

Test old session fixtures.

NOW · NEXTINVESTIGATION

Use F-ADAPTER for subsequent evaluation.

BEFORE · STATUS

preliminary

NOW · STATUS

contradicted

BEFORE · TITLE

Session compatibility hypothesis

NOW · TITLE

Namespace-only session hypothesis contradicted

BEFORE · UNCERTAINTIES

["No deserialization evidence"]

NOW · UNCERTAINTIES

["Legacy session diversity unknown"]

evidence / E-DEPS2 · revision — → 1

added

Complete synthetic direct inventory with two interface smoke checks.

New record. Its complete content is included in the cumulative report.

evidence / E-MODEL2 · revision — → 1

added

Revised synthetic effort model incorporating session adapter work.

New record. Its complete content is included in the cumulative report.

evidence / E-SERIAL · revision — → 1

added

Synthetic legacy fixture experiment: 9/32 fail without adapter; 32/32 pass with adapter.

New record. Its complete content is included in the cumulative report.

evidence / E-TX · revision — → 1

added

Synthetic transaction rollback experiment: 18/18 passed on a representative slice.

New record. Its complete content is included in the cumulative report.

risks / R-SESSION · revision — → 1

added

Stored session incompatibility raises adaptation effort and backward-compatibility uncertainty.

New record. Its complete content is included in the cumulative report.

questions / Q-SERIAL · revision 1 → 2

resolved

Will legacy session fixtures deserialize safely?

▶ 3 changed fields · inspect previous and current understanding

questions / Q-SUPPORT · revision — → 1

added

Which support terms and security policy oracle are authoritative?

New record. Its complete content is included in the cumulative report.

estimates / EST-EFFORT · revision 1 → 2

revised

Technical adaptation effort

▶ 5 changed fields · inspect previous and current understanding

gates / G-COMPAT · revision 1 → 2

revised

Representative compatibility evidence

▶ 3 changed fields · inspect previous and current understanding

gates / G-DEPS · revision 1 → 2

revised

Maintained dependency path demonstrated

▶ 3 changed fields · inspect previous and current understanding

gates / G-OPS · revision 1 → 2

revised

Operational recovery assessment

▶ 2 changed fields · inspect previous and current understanding

decisions / DEC-COUNTERFACTUAL · revision — → 1

added

Agree an economic counterfactual and horizon

New record. Its complete content is included in the cumulative report.

decisions / DEC-EXPERIMENT · revision 1 → 2

revised

Authorize two assessment experiments

▶ 3 changed fields · inspect previous and current understanding

recommendations / REC-ATLAS · revision 1 → 2

revised

Defer migration commitment; continue focused assessment.

▶ 6 changed fields · inspect previous and current understanding

activities / ACT-RESEARCH · revision 1 → 2

revised

Serialization and transaction research experiments

▶ 3 changed fields · inspect previous and current understanding

executiveSummary / summary · revision — → —

revised

Executive synthesis updated

▶ 1 changed fields · inspect previous and current understanding

Decision Readiness

Decision evidence gates

2/4 met (50%). Critical gates are binding; readiness is not an approval score.

✓ G-COMPAT · CRITICAL

Representative compatibility evidence met

Sampled API and serialization behavior is evidenced; this gate is limited to representative evidence.

□ E-TX · validated □ E-SERIAL · validated

✓ G-DEPS · ADVISORY

Maintained dependency path demonstrated met

A representative maintained replacement path is documented; complete legal clearance is tracked separately.

□ E-DEPS2 · reviewed

? G-ECON · CRITICAL

Defensible worthiness model unknown

Benefit model and counterfactual missing.

□ E-MODEL · reviewed

! G-OPS · CRITICAL

Operational recovery assessment unmet

Investigation started, but no restore/replay evidence is available.

Unverified · no supporting evidence

Recommendation & limitations

Defer migration commitment; continue focused assessment.

Technical feasibility is increasingly evidenced, but worthiness is not established without the economic counterfactual and recovery assessment.

- Unknown benefits
- No recovery evidence
- License exception unresolved

□ F-API

□ F-ADAPTER

□ F-DEPS

> Recommendation status: recommended. Report approval records the reviewed assessment; it does not approve migration execution.

Outstanding management decisions

DECISION	STATUS	OWNER / DUE	RATIONALE / PROVENANCE
DEC-COUNTERFACTUAL **Agree an economic counterfactual and horizon**	proposed	Finance sponsor 2026-11-16	Worthiness cannot be evaluated without a baseline and horizon. □ E-MODEL2 · reviewed
DEC-EXPERIMENT **Authorize two assessment experiments**	approved	Architecture sponsor 2026-11-06	Assessment experiment funding approved. This does not approve migration execution. □ E-TX · validated □ E-SERIAL · validated

Next assessment activities

RESEARCH ACTIVITY	STATUS	OWNER / NEXT ACTION
ACT-RESEARCH Serialization and transaction research experiments	completed	Research agents + reviewer Investigate operational recovery and broaden fixtures. □ E-TX · validated □ E-SERIAL · validated

Technical Appendix

Approved reporting history

2026-W45

2026-11-02 · Demonstration reviewer
`726c6084703d0e3614f1…`

2026-W46

2026-11-09 · Demonstration reviewer
`8339db9fa5ba607921d0…`

Publication & provenance contract

Schema / presentation

1.0 / 1.0

Reviewed by

Demonstration reviewer

Approval timestamp

2026-11-09T14:46:04.552682+00:00

Last update actor

synthetic-research-agent

Update method

ingest

> The HTML is generated by Quarto from schema-validated content and fixed components. No LLM executes during ingestion or rendering.

Historical comparisons are computed using stable IDs. “Strengthened” and “weakened” reflect explicit confidence changes; a text-only conclusion change is “revised.”

Original source material

SOURCE	INTEGRITY	ACCESS
01-assessment-charter.md 171 lines · original Markdown retained	`fa3163929480aa01a20f4d89…`	
02-technical-analysis.md 151 lines · original Markdown retained	`90f89252abc6b201fdf5cb15…`	
03-economics-and-decisions.md 159 lines · original Markdown retained	`d4ebea1d341c46b82d7406d7…`	
04-runtime-validation.md 211 lines · original Markdown retained	`4836c587a0824ecc17a347f2…`	
05-dependency-and-security-review.md 70 lines · original Markdown retained	`fa36424386f9d6d082e9c380…`	
06-worthiness-reassessment.md 96 lines · original Markdown retained	`d0f46f3e79fa31fe4ee5194d…`	

Measurement definitions

Analysis coverage = Σ(dimension weight × examined investigation weight / total planned investigation weight) / Σ(dimension weight). Examined: preliminary, corroborated, validated, inconclusive, contradicted, superseded. Started and not-examined investigations contribute zero. Any dimension without planned scope makes overall coverage Unknown.

Evidence validation = validated evidence / all evidence. Finding maturity = corroborated-or-validated findings / all findings. Open uncertainties = questions explicitly marked open; deferred questions remain separately visible. Decision gates met = met / all gates. No metrics imply implementation completion.

Assessment Intelligence · 2026-11-09 · Research maturity, not migration delivery

`Approved state SHA-256 8339db9fa5ba607921d075c5140cf2867b50d19aecb2254a685189501397d574`